FIT3003 Business Intelligence and Data Warehousing, Assignment 2

Should MonCity invest more in its self driving cars?

Ishan Sharma, 34670068. Data warehouse analysis of the MonCity self driving car project, 2015 to 2021.

Seven years of MonCity booking, maintenance and accident records were put into a data warehouse to look at how the cars are used, what they cost to run and how safe they are.

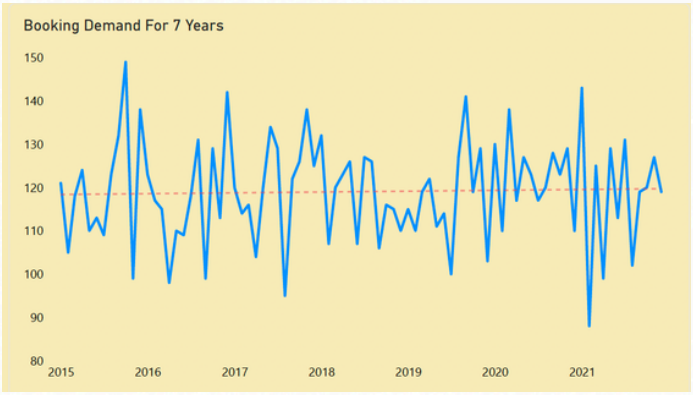
How it was done

1. Exploring the data. Each table was checked for gaps, duplicates and odd values.
2. Cleaning. 6 problems were fixed, like a duplicate booking and a negative cost.
3. Building the warehouse. Three fact tables were made for bookings, maintenance and accidents.
4. Analysis. SQL queries were run and a Power BI dashboard was made.

10,000 bookings over 7 years. **30 cars, 10 of each type.** **$302,700 total maintenance cost.** 992 accidents, close to 1 in every 10 bookings.

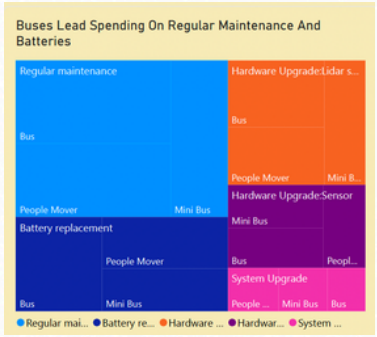
1. Demand is not going up

It was found that bookings stayed at around 119 a month across the seven years. The trend line only went up by about 1% over the whole time, so nothing shows that more cars are needed.



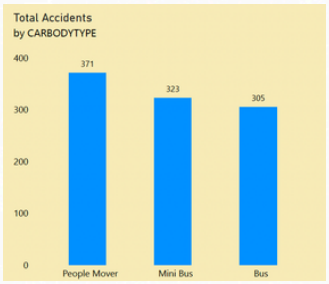
2. Buses cost the most to maintain

Each car type had nearly the same number of bookings, around 330 per car. But one bus was found to cost about $2,490 more to maintain than one mini bus.



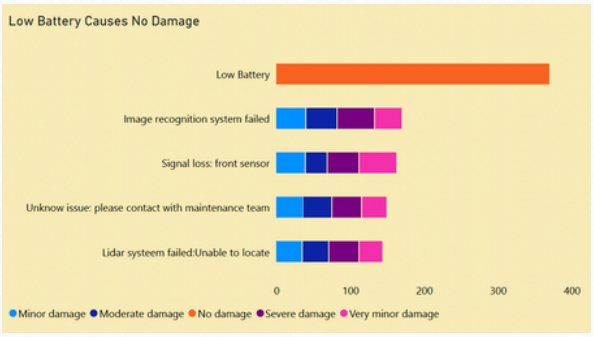
People movers have the most accidents

People movers were involved in 22% more accidents than buses. Mini buses were in the middle for accidents and the cheapest to maintain, so they were seen as the best overall.



3. Sensor problems cause the damage, not low battery

Low battery was the most common error, but none of those accidents caused any damage. All accidents with damage were linked to the sensors or an unknown error. Image recognition failure alone caused 51 severe accidents.



Recommendation

1. No new cars should be bought. Bookings are not growing and the fleet already has enough room.
2. The sensors should be fixed first, mainly image recognition, since it is linked to the most severe accidents.
3. Batteries should be managed better, for example with set charging times, to avoid the 369 low battery cases.
4. Old buses could be swapped for mini buses, since they handle the same demand for less money.

Why not invest more

There is close to 1 accident for every 10 trips, and newer cars were found to have about 20% more accidents than older ones. Because of this, more money is not worth it until the sensor issues are sorted out.